Erdős 252

Irrationality of the factorial divisor-sum series

Tokengrinder

Complete mathematical exposition of the Lean proof

Abstract

For every natural number k, the series $\sum_{n\geq 1} \sigma_k(n)/n!$ is irrational. Rationality would make its factorial-scaled tails eventually integral. A fixed finite grid of pairwise coprime dilations cancels the lower orders in an exact Stirling expansion. The remaining divisor-phase sum would tend to zero on an arithmetic progression, whereas two fresh-prime refinements of its progression mean give incompatible limits. Degree zero is settled separately by positivity and decay of the actual tail.

This document states and proves every named theorem and lemma, including private lemmas, in `Erdos252/Solution.lean`, and records every local definition. The final section also covers all six statement-audit theorems. The typewritten identifier below each lemma title is its exact Lean name in the `Erdos252` namespace, except in the audit section. Mathlib results used as standard prerequisites are identified in the text; the library itself is not reproduced here. The Lean source, not this exposition, is the kernel-checked artifact.

Contents

1 Conventions and the argument

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$. Unless specified otherwise, indices and exponents are natural numbers. Put $\sigma_k(0) = 0$ and, for $n > 0$,

$$\sigma_k(n) = \sum_{d|n} d^k, \qquad \alpha_k = \sum_{n=0}^{\infty} \frac{\sigma_k(n)}{n!}.$$

The theorem is that $\alpha_k \notin \mathbb{Q}$ for every $k \in \mathbb{N}$. The ascending factorial is $n^{\overline{h}} = \prod_{j<h}(n+j)$, with empty product 1. Natural subtraction is truncated: $a \dot{-} b = \max(a - b, 0)$. The integer square root is $\lfloor \sqrt{n} \rfloor$.

To preserve the full scope of the auxiliary Lean statements, division in $\mathbb{R}$ is totalized by $a/0 = 0$, and $0^0 = 1$. Natural division by zero is also zero. A primed infinite sum $\sum'$ denotes Lean's `tsum`: the ordinary sum when the family is summable, and zero otherwise. Every infinite sum used for the final contradiction is proved summable; this convention only matters for auxiliary identities at degenerate parameters. Saying that a family has sum s asserts convergence to s, not merely a totalized-sum identity. All sequence limits are as the natural index tends to infinity. An *eventual* statement holds at every sufficiently large index. Finite-type sums include every element.

First, we obtain an exact finite expansion of the factorial tail with error $o(1/n)$. Next, finite differences on a fixed coprime dilation grid cancel the unwanted orders. Finally, actual progression means of the surviving divisor phases separate one shift using a fresh prime. No prime-pattern conjecture is involved. The estimates also prove $k = 0$ directly.

The central estimates are Lemmas 4.8 and 5.10. The cancellation step is Lemma 9.4; the incompatible limits are Lemmas 10.11 and 11.1. The complete conclusion is Lemma 11.6.

2 Elementary bounds and integrality

Lemma 2.1 (Convergence of the original series)**.**

`summable_sigma_factorial`

For every k, the family $\sigma_k(n)/n!$, $n \in \mathbb{N}$, is summable.

Proof. Mathlib's divisor bound gives $\sigma_k(n) \le n^{k+1}$. Since $n \le 2^n$, we have $0 \le \sigma_k(n)/n! \le (2^{k+1})^n/n!$. Comparison with this exponential series proves summability. □

Lemma 2.2 (An integral null sequence)**.**

`eventually_zero_of_int`

If $f : \mathbb{N} \to \mathbb{R}$ tends to zero and is eventually integer-valued, then $f(n) = 0$ eventually.

Proof. Eventually $|f(n)| < 1$, and the only integer in that interval is zero. Intersect this eventual condition with eventual integrality. □

Lemma 2.3 (Pairing divisors)**.**

`card_divisors_le_sqrt`

For $n > 0$, the number of positive divisors of n is at most $2\lfloor \sqrt{n} \rfloor$.

Proof. Let $S = \{1, \ldots, \lfloor \sqrt{n} \rfloor\}$. Every divisor d has $d \in S$ or $n/d \in S$, since $d(n/d) = n$. In the latter case $d = n/(n/d)$. The divisor set is therefore contained in $S \cup \{n/s : s \in S\}$, of cardinality at most $2|S|$. □

Lemma 2.4 (A square-root divisor bound)**.**

`sigma_le_pow_sqrt`

For all k and $n > 0$, $\sigma_k(n) \le 64 n^k \sqrt{n}$.

Proof. Each divisor contributes at most n^k. The preceding count and $\lfloor \sqrt{n} \rfloor \le \sqrt{n}$ give $2n^k\sqrt{n}$. The weaker constant 64 is retained throughout the formalization. ☐

Lemma 2.5 (Sublinear square-root growth)**.**

`tendsto_sqrt_div_nat`

$\sqrt{n}/n \longrightarrow 0$.

Proof. For $n > 0$, $\sqrt{n}/n = 1/\sqrt{n}$, and $\sqrt{n} \to \infty$. ☐

3 Reciprocal products and Stirling remainders

The definitions `descRecip`, `stirlingPoly`, and `stirlingErr` are, for $x \in \mathbb{R}$,

$$D_h(x) = \frac{1}{\prod_{a<h}(x-a)}, \qquad P_{j,L}(x) = \sum_{i<L} \frac{\left\{ {i \atop j} \right\}}{x^{i+1}}, \qquad E_{j,L}(x) = D_{j+1}(x) - P_{j,L}(x).$$

The coefficients are Stirling numbers of the second kind. These definitions do not assume nonzero denominators.

Lemma 3.1 (Adding a descending factor)**.**

`descRecip_succ`

$D_{h+1}(x) = D_h(x)/(x-h)$ for every h and $x \in \mathbb{R}$.

Proof. Split the final product factor and use $1/(uv) = (1/u)/v$. ☐

Lemma 3.2 (The first reciprocal product)**.**

`descRecip_one`

$D_1(x) = 1/x$.

Proof. The defining product consists of the single factor x. ☐

Lemma 3.3 (Finite coefficient recurrence)**.**

`stirlingPoly_recurrence`

For all j, L and $x \in \mathbb{R}$,

$$P_{j+1,L+1}(x) = \frac{P_{j,L}(x) + (j+1)P_{j+1,L}(x)}{x}.$$

Proof. The zeroth term on the left is zero. For each remaining term apply $\left\{ {i+1 \atop j+1} \right\} = \left\{ {i \atop j} \right\} + (j+1)\left\{ {i \atop j+1} \right\}$, then distribute the finite sums and the extra factor $1/x$. ☐

Lemma 3.4 (Exact error recurrence)**.**

`stirlingErr_recurrence`

If $x \neq 0$ and $x - (j+1) \neq 0$, then

$$E_{j+1,L+1}(x) = \frac{E_{j,L}(x) + (j+1)E_{j+1,L}(x)}{x}.$$

Proof. The reciprocal recurrence gives $xD_{j+2} = D_{j+1} + (j+1)D_{j+2}$ by clearing the stated nonzero denominators. Subtract the polynomial recurrence. This remains valid if another product factor vanishes, by the totalized-division convention. ☐

Lemma 3.5 (Bounds for a descending reciprocal)**.**

`descRecip_bounds`

If $1 \le h \le H \le x$, *then* $0 < D_h(x) \le 1/x$.

Proof. Induct on $h \ge 1$. Initially $D_1 = 1/x > 0$. In the step, $x - h \ge 1$, so dividing the preceding positive reciprocal by $x - h$ preserves positivity and does not increase it. ☐

Lemma 3.6 (Uniform nonnegative error bound)**.**

`stirlingErr_bounds`

If $j + 1 \le H \le x$, *then for every* L, $0 \le E_{j,L}(x) \le H^L/x^{L+1}$.

Proof. Here $x > 0$. Induct on L, simultaneously for all admissible j. At $L = 0$ the polynomial is empty, so use the preceding bound. At the next order, $j = 0$ gives zero error, because $\left\{ {0 \atop 0} \right\} = 1$ and $\left\{ {i+1 \atop 0} \right\} = 0$ make $P_{0,L+1}(x) = 1/x$. For the index $j + 1$ put $u = H^L/x^{L+1} \ge 0$. The two errors in the recurrence lie in $[0, u]$, and $j + 2 \le H$. Their nonnegative weighted sum is at most $u + (j+1)u = (j+2)u \le Hu$. Dividing by x gives H^{L+1}/x^{L+2}. ☐

Lemma 3.7 (Centering at the largest factor)**.**

`descRecip_centered`

For all n, h, $D_h(n + h) = 1/(n + 1)^{\overline{h}}$.

Proof. The descending product $(n+h) \cdots (n+1)$ equals the ascending factorial. Lean uses Mathlib's descending/ascending factorial identity; all natural subtractions in its product are exact. ☐

4 The actual factorial tail

The definitions `alpha`, `seriesPrefix`, and `scaledTail` are

$$\alpha_k = \sum_{m \ge 0} \frac{\sigma_k(m)}{m!}, \qquad U_k(n) = \sum_{m<n} \frac{\sigma_k(m)}{m!}, \qquad T_k(n) = (n \mathbin{\dot{-}} 1)!\,[\alpha_k - U_k(n)].$$

For $n > 0$, this tail starts at term n, not at $n+1$. The definitions `blockTerm`, `tailMain`, `tailErr`, `omittedTail`, and `finiteErr` are

$$\begin{aligned}
B_k(n, j) &= \frac{\sigma_k(n+j)}{n^{\overline{j+1}}}, \\
M_k(n) &= \sum_{j=0}^{k} \sigma_k(n+j+1) P_{j,k+1}(n+j+1), \\
R_k(n) &= T_k(n+1) - M_k(n), \\
O_k(n, H) &= {\sum_{r \in \mathbb{N}}}' B_k(n+1, r+H), \\
F_k(n) &= \sum_{j=0}^{k} \sigma_k(n+j+1) E_{j,k+1}(n+j+1).
\end{aligned}$$

Here R_k is the tail error; the later survivor is denoted V_k.

Lemma 4.1 (Integrality of the scaled prefix).

`seriesPrefix_integral`

For every k, n, $(n \dot{-} 1)!\, U_k(n) \in \mathbb{Z}$.

Proof. For $m < n$, $m! \mid (n-1)!$. The scaled sum is the natural number $\sum_{m<n}((n-1)!/m!)\sigma_k(m)$. For $n = 0$ the sum is empty. ☐

Lemma 4.2 (Late integral tails).

`eventually_scaledTail_integral`

If $\alpha_k \in \mathbb{Q}$, then $T_k(n) \in \mathbb{Z}$ eventually.

Proof. Write $\alpha_k = a/b$ with $a \in \mathbb{Z}$ and $b \geq 1$. If $n \geq b+1$, then $b \mid (n-1)!$, so $(n-1)!\alpha_k$ is integral. Subtract the integral scaled prefix. ☐

Lemma 4.3 (The actual block summand).

`scaled_summand_eq_blockTerm`

For $n > 0$, $(n-1)!\dfrac{\sigma_k(j+n)}{(j+n)!} = B_k(n, j)$.

Proof. Commute $j + n$, use $(n+j)! = (n-1)!\, n^{\overline{j+1}}$, and cancel the nonzero factorial $(n-1)!$. ☐

Lemma 4.4 (The actual infinite block tail).

`scaledTail_tsum`

For $n > 0$, $T_k(n) = \sum'_{j \in \mathbb{N}} B_k(n, j)$.

Proof. Split the summable original series at n, subtract its finite prefix, and distribute $(n-1)!$ through the tail. Identify each summand by the preceding lemma. ☐

Lemma 4.5 (Summability of block terms).

`summable_blockTerm`

For $n > 0$, the family $j \mapsto B_k(n, j)$ is summable.

Proof. It is the shifted original series multiplied by $(n-1)!$. ☐

Lemma 4.6 (Nonnegative block terms).

`blockTerm_nonneg`

$B_k(n, j) \geq 0$ for every k, n, j.

Proof. It is the totalized quotient of two nonnegative reals. ☐

Lemma 4.7 (Nonnegative omitted tail).

`omittedTail_nonneg`

$O_k(n, H) \geq 0$ for every k, n, H.

Proof. Sum the nonnegative block terms. ☐

Lemma 4.8 (Exact decomposition of the error).

`tailErr_eq`

For every k, n, $R_k(n) = O_k(n, k+1) + F_k(n)$.

Proof. Split the summable series for $T_k(n+1)$ after $k+1$ terms. Centering the reciprocal product gives $B_k(n+1,j) = \sigma_k(n+j+1)D_{j+1}(n+j+1)$. Subtract $M_k(n)$ and distribute subtraction through the finite sum. The finite differences are exactly $F_k(n)$; the omitted tail remains. This is the direct decomposition used in the compressed Lean proof. $\square$

Lemma 4.9 (The exact main-plus-error identity)**.**

`scaledTail_expansion`

$T_k(n+1) = M_k(n) + R_k(n)$ for every k, n.

Proof. Rearrange the definition of $R_k(n)$. $\square$

Lemma 4.10 (The expanded finite main term)**.**

`tailMain_eq_range`

For every k, n,

$$M_k(n) = \sum_{j=0}^{k}\sum_{i=0}^{k} \left\{ {i \atop j} \right\} \frac{\sigma_k(n+j+1)}{(n+j+1)^{i+1}}.$$

Proof. Insert the sum defining $P_{j,k+1}$ and distribute $\sigma_k(n+j+1)$. $\square$

5 Quantitative tail estimates

Lemma 5.1 (Polynomial versus ascending factorial)**.**

`poly_ascending_bound`

For $d > 0$ and all n, r,

$$(n+1+r+d)^d(n+1)^{r+1} \le d^d(n+1)^{\overline{r+d+1}}.$$

Proof. Since $d \ge 1$, $n+1+r+d \le d(n+1+r+1)$. Each factor of an ascending factorial is at least its first factor. Raise the first inequality to power d, then bound the resulting powers by $(n+1+r+1)^{\overline{d}}$ and $(n+1)^{\overline{r+1}}$. Their product is $(n+1)^{\overline{r+d+1}}$. $\square$

Lemma 5.2 (A reciprocal polynomial block bound)**.**

`poly_block_le`

For $d > 0$,

$$\frac{(n+1+r+d)^d}{(n+1)^{\overline{r+d+1}}} \le \frac{d^d}{(n+1)^{r+1}}.$$

Proof. Both denominators are positive. Cross-multiplication reduces the inequality to the preceding lemma. $\square$

Lemma 5.3 (A uniform numerator bound)**.**

`sigma_le_pow_succ_div_sqrt`

If $n+1 \le m$, then

$$\sigma_k(m) \le \frac{64}{\sqrt{n+1}}\, m^{k+1}.$$

Proof. Here $m > 0$. The square-root divisor bound gives $64m^k\sqrt{m} = 64m^{k+1}/\sqrt{m}$. Use $\sqrt{m} \ge \sqrt{n+1} > 0$. $\square$

Lemma 5.4 (Geometric domination of each omitted term).

`omittedTerm_le`

For all k, n, r,

$$B_k(n+1, r+k+1) \le \frac{64}{\sqrt{n+1}} \frac{(k+1)^{k+1}}{(n+1)^{r+1}}.$$

Proof. Apply the preceding numerator bound with $m = n+1+r+k+1$, divide by its positive ascending factorial, and apply the polynomial block bound with $d = k+1$. □

Lemma 5.5 (The geometric majorant).

`hasSum_geometric_recip`

For $C \in \mathbb{R}$ and $n > 0$, the family $C/(n+1)^{r+1}$, $r \in \mathbb{N}$, has sum C/n.

Proof. Its ratio is $q = 1/(n+1) \in (0,1)$ and its initial term is $C/(n+1)$. The geometric-series formula gives $(C/(n+1))/(1-q) = C/n$. □

Lemma 5.6 (The omitted-tail bound).

`omittedTail_le`

For $n > 0$,

$$O_k(n, k+1) \le \frac{64(k+1)^{k+1}}{n\sqrt{n+1}}.$$

Proof. The omitted block family is a tail of a summable block family. Compare it term by term with the preceding geometric majorant, using $C = 64(k+1)^{k+1}/\sqrt{n+1}$, and sum. □

Lemma 5.7 (The scaled omitted-tail bound).

`omittedTail_scaled_le`

For $n > 0$,

$$(n+1)O_k(n, k+1) \le \frac{128(k+1)^{k+1}}{\sqrt{n+1}}.$$

Proof. Multiply the preceding bound by $n+1$ and use $(n+1)/n \le 2$. □

Lemma 5.8 (One finite denominator remainder).

`finiteErr_term_bound`

If $n \ge k+1$ and $0 \le j \le k$, then

$$0 \le \sigma_k(n+j+1)E_{j,k+1}(n+j+1) \le \frac{64(k+1)^{k+1}\sqrt{n+1}}{(n+1)^2}.$$

Proof. Put $x = n+j+1$. Then $x \ge n+1 > 0$ and $x \ge k+1$. Reuse the numerator estimate from the omitted-tail argument and the Stirling error bound with $H = L = k+1$:

$$0 \le \sigma_k(x)E_{j,k+1}(x) \le \frac{64x^{k+1}}{\sqrt{n+1}} \frac{(k+1)^{k+1}}{x^{k+2}} = \frac{64(k+1)^{k+1}}{x\sqrt{n+1}}.$$

Since $x \ge n+1$, replace x in the denominator by $n+1$ and use $1/\sqrt{n+1} = \sqrt{n+1}/(n+1)$. No upper bound on $\sqrt{x}$ is needed. □

Lemma 5.9 (The complete finite error).

`finiteErr_bounds`

If $n \ge k+1$, then $F_k(n) \ge 0$ and

$$(n+1)F_k(n) \le 64(k+1)^{k+2}\frac{\sqrt{n+1}}{n+1}.$$

Proof. Sum the preceding nonnegative bounds over the $k+1$ indices $j = 0, \dots, k$, then multiply by $n+1$ and cancel one power. $\square$

Lemma 5.10 (The exact error is $o(1/n)$)**.**

`tendsto_tailErr_mul`

For every k, $(n+1)R_k(n) \longrightarrow 0$.

Proof. The scaled finite error is eventually squeezed between zero and a fixed multiple of $\sqrt{n+1}/(n+1) \to 0$. The scaled omitted tail is squeezed between zero and a fixed multiple of $1/\sqrt{n+1} = \sqrt{n+1}/(n+1) \to 0$. Add these two limits using the exact decomposition of R_k. $\square$

Lemma 5.11 (Decay of the unscaled error)**.**

`tendsto_tailErr`

For every k, $R_k(n) \longrightarrow 0$.

Proof. Divide the preceding null sequence by $n+1 \to \infty$. $\square$

6 Divisor phases on arithmetic progressions

Lemma 6.1 (Solving a coprime affine congruence)**.**

`affine_exists_residue`

If $d > 0$ and $\gcd(Q, d) = 1$, there exists $v < d$ such that $d \mid Qv + A$.

Proof. In $\mathbb{Z}/d\mathbb{Z}$, Q is invertible. Choose the representative in $\{0, \dots, d-1\}$ of $-AQ^{-1}$. $\square$

Lemma 6.2 (The complete affine solution class)**.**

`affine_iff_modEq`

If $\gcd(Q, d) = 1$ and $d \mid Qv + A$, then for every j,

$$d \mid Qj + A \iff j \equiv v \pmod{d}.$$

Proof. Compare $Qj+A$ with $Qv+A$ modulo d, cancel A, and cancel the coprime multiplier Q. The reverse implication follows by multiplying the congruence by Q and adding A. This formulation also permits $d = 0$, where congruence means equality. $\square$

Define `phase`, `divTerm`, `meanTerm`, and `progMean` by

$$\rho_k(n) = \frac{\sigma_k(n)}{n^k}, \qquad b_k(d, n) = \frac{\mathbf{1}_{\{d|n\}}}{d^k},$$
$$m_k(Q, A, d) = \mathbf{1}_{\{\gcd(d,Q)|A\}} \frac{\gcd(d, Q)}{d^{k+1}}, \qquad \mu_k(Q, A) = \sum_{d\in\mathbb{N}}{}' m_k(Q, A, d).$$

The indicator is one when its condition holds and zero otherwise. These definitions include $d = 0$ and $n = 0$ with the conventions of Section 1.

Lemma 6.3 (The reciprocal-divisor representation)**.**

`hasSum_divTerm`

For $n > 0$ and every k, the family $d \mapsto b_k(d, n)$ has sum $\rho_k(n)$.

Proof. Pair each positive divisor with its complement: $\sigma_k(n) = \sum_{d|n}(n/d)^k$. After division by $n^k > 0$, each term is d^{-k}. The family $b_k(d, n)$ vanishes outside the finite positive divisor set, so this finite equality gives the asserted convergent infinite sum. $\square$

Lemma 6.4 (Averages count affine solutions)**.**

`divTerm_sum_eq_count`

For every k, Q, A, d, N, let $C_d(N) = |\{j < N : d \mid Qj + A\}|$. Then

$$\frac{1}{N}\sum_{j<N} b_k(d, Qj + A) = \frac{C_d(N)}{N\,d^k}.$$

Proof. Every selected index contributes the same number $1/d^k$. Sum this constant over the filtered finite set, then divide by N. ☐

Lemma 6.5 (Density of one residue class)**.**

`tendsto_count_modEq_div`

For $r > 0$ and every v,

$$\frac{|\{j < N : j \equiv v \pmod r\}|}{N} \longrightarrow \frac{1}{r}.$$

Proof. The count is $\lfloor N/r\rfloor + \mathbf{1}_{\{v \bmod r < N \bmod r\}}$. The first term divided by N tends to $1/r$: the difference between N/r and $\lfloor N/r\rfloor$ lies in $[0, 1)$, and Lean uses the corresponding Mathlib limit for a normalized floor. The second term divided by N is between zero and $1/N$ and tends to zero. ☐

Lemma 6.6 (The mean of one divisor term)**.**

`divTerm_cesaro`

For $k > 0$ and all Q, A, d,

$$\frac{1}{N}\sum_{j<N} b_k(d, Qj + A) \longrightarrow m_k(Q, A, d).$$

Proof. For $d = 0$ both sides vanish. Otherwise let $g = \gcd(d, Q) > 0$. If $g \nmid A$, there are no affine solutions: g divides d and Qj, so $d \mid Qj + A$ would force $g \mid A$. If $g \mid A$, then $\gcd(Q/g, d/g) = 1$, and the two affine congruence lemmas solve the reduced congruence with offset A/g; multiplying by g, which divides d, Q and A, shows that the solutions form one residue class modulo d/g. Its density is g/d; multiply by d^{-k} to obtain g/d^{k+1}. ☐

Lemma 6.7 (Bounding the affine count)**.**

`affine_count_le_quotient`

If $Q > 0$ and $A > 0$, then for all d, N,

$$C_d(N) \le \left\lfloor \frac{QN + A}{d} \right\rfloor.$$

Proof. The injective map $j \mapsto Qj + A$ sends counted indices into the positive multiples of d at most $QN + A$. There are exactly the displayed number of such multiples. For $d = 0$ both sets are empty. ☐

Lemma 6.8 (Summable domination of the averages)**.**

`divTerm_cesaro_bound`

If $k, Q, A > 0$, then for every d, N,

$$\left|\frac{1}{N}\sum_{j<N} b_k(d, Qj + A)\right| \le \frac{Q + A}{d^{k+1}}.$$

Proof. If $N = 0$ or $d = 0$, the average is zero. Otherwise all terms are nonnegative, and the counting bound gives $C_d(N)d \le QN + A \le (Q + A)N$, since $N \ge 1$. Divide by $Nd^{k+1} > 0$ and use the exact counting formula. ☐

Lemma 6.9 (Bounds for a local gcd mean).

`meanTerm_bounds`

For $Q > 0$ and all k, A, d, $0 \le m_k(Q, A, d) \le Q/d^{k+1}$.

Proof. The incompatible case is zero; in the compatible case use $0 \le \gcd(d, Q) \le Q$ and divide by the nonnegative denominator. The $d = 0$ term is also zero. ☐

Lemma 6.10 (Summability of the progression mean).

`summable_meanTerm`

For $k, Q > 0$ and every A, $d \mapsto m_k(Q, A, d)$ is summable.

Proof. The preceding comparison is dominated by the convergent p-series Q/d^{k+1}, since $k + 1 > 1$. ☐

Lemma 6.11 (The actual progression mean).

`tendsto_progMean`

For $k, Q, A > 0$,

$$\frac{1}{N} \sum_{j<N} \rho_k(Qj + A) \longrightarrow \mu_k(Q, A).$$

Proof. For each d the averaged reciprocal-divisor term converges to $m_k(Q, A, d)$. Its absolute value is bounded, uniformly in N, by $(Q + A)/d^{k+1}$, a summable family. Mathlib's dominated-convergence theorem for series therefore permits summing these limits over d. For each fixed N, interchange the finite average with the convergent divisor sums, using the reciprocal-divisor representation. This gives the displayed phase average. In particular the argument works for $k = 1$: domination is on the averages, with exponent 2, not on pointwise divisor terms by a divergent harmonic series. ☐

7 Fresh-prime refinements and the isolated shift

Lemma 7.1 (Two separating prime residues).

`fresh_prime_residues`

Let I be a finite type, $i_0 \in I$, $r_i > 0$, $Q > 0$, and $A \in \mathbb{N}$. There exist a prime L coprime to Q and $v_0, v_1 \in \mathbb{N}$ such that

$$\begin{aligned} &L \nmid Qv_0 + A + r_i \quad \textit{for every } i, \\ &L \mid Qv_1 + A + r_i \quad \iff \quad r_i = r_{i_0}. \end{aligned}$$

Proof. By the infinitude of primes, choose $L > Q$ and $L > \max_i r_i$. Then $\gcd(Q, L) = 1$. Choose v_0 solving $Qv_0 + A \equiv 0 \pmod{L}$ and v_1 solving $Qv_1 + A + r_{i_0} \equiv 0 \pmod{L}$. The first residue misses all shifts because $0 < r_i < L$. For the second, divisibility is equivalent to $r_i \equiv r_{i_0} \pmod{L}$, which is equality because both representatives lie strictly between zero and L. ☐

Lemma 7.2 (Finite weighted averages).

`tendsto_weightedCesaro`

Let I be a finite type, $f_i : \mathbb{N} \to \mathbb{R}$, and $c_i, \mu_i \in \mathbb{R}$. If $N^{-1} \sum_{n<N} f_i(n) \to \mu_i$ for every i, then

$$\frac{1}{N} \sum_{n<N} \sum_{i \in I} c_i f_i(n) \longrightarrow \sum_{i \in I} c_i \mu_i.$$

Proof. Commute the finite sums and move the fixed coefficients outside. A finite sum of convergent sequences converges to the sum of its limits. ☐

Lemma 7.3 (The prime gcd dichotomy).

`gcd_eq_prime_or_one`

If L is prime, then $\gcd(d, L) = L$ *when* $L \mid d$, *and* $\gcd(d, L) = 1$ *otherwise.*

Proof. In the first case the gcd is L. In the second, primality implies d and L are coprime. $\square$

Lemma 7.4 (Multiplying the divisor index).

`meanTerm_mul`

For all k, Q, A, d, L *with* $\gcd(L, Q) = 1$,

$$m_k(Q, A, Ld) = \frac{m_k(Q, A, d)}{L^{k+1}}.$$

Proof. Coprime cancellation gives $\gcd(Ld, Q) = \gcd(d, Q)$. The divisibility indicator is unchanged, and $(Ld)^{k+1} = L^{k+1}d^{k+1}$. If $L = 0$, both sides are zero. $\square$

Lemma 7.5 (The mean supported on multiples).

`progMean_multiples`

For all k, Q, A *and* $L > 0$ *with* $\gcd(L, Q) = 1$,

$$\sum_{d\in\mathbb{N}}{}' \mathbf{1}_{\{L|d\}}\, m_k(Q, A, d) = \frac{\mu_k(Q, A)}{L^{k+1}}.$$

Here L need not be prime.

Proof. The summand is supported on multiples of L. Reindex them injectively as $d = Le$, then use the preceding identity and pull the fixed scalar outside the sum. The equality is also valid for nonsummable families with the totalized-sum convention: reindexing and multiplication by the nonzero scalar $L^{-(k+1)}$ preserve summability or nonsummability. $\square$

Lemma 7.6 (Refining one local gcd mean).

`meanTerm_refine`

If L is prime and $\gcd(L, Q) = 1$, *then for every* k, B, d,

$$\begin{aligned} m_k(QL, B, d) = m_k&(Q, B, d) \\ &+ \left((L-1)\mathbf{1}_{\{L|B\}} - \mathbf{1}_{\{L\nmid B\}}\right)\mathbf{1}_{\{L|d\}}\, m_k(Q, B, d). \end{aligned}$$

Proof. Write $g = \gcd(d, Q)$. If $L \nmid d$, the refined gcd is still g and the correction vanishes. If $L \mid d$, the refined gcd is gL. Since $\gcd(g, L) = 1$, $gL \mid B$ is equivalent to $g \mid B$ and $L \mid B$. Thus the refined term is L times the original term when $L \mid B$, and zero otherwise. These cases give the formula. $\square$

Lemma 7.7 (The exact prime-refinement factor).

`progMean_refine`

If $k, Q > 0$, *L is prime,* $\gcd(L, Q) = 1$, *and* $B \equiv A \pmod{Q}$, *then*

$$\mu_k(QL, B) = \mu_k(Q, A)\left(1 - L^{-(k+1)} + \mathbf{1}_{\{L|B\}}L^{-k}\right).$$

Proof. The local-mean family and its restriction to multiples of L are summable. Sum the local refinement identity, substitute the multiples formula, and use $\mu_k(Q, B) = \mu_k(Q, A)$, which holds because each $\gcd(d, Q)$ divides Q, so it divides B if and only if it divides A. When $L \mid B$, the multiplier is $1 + (L-1)L^{-(k+1)}$; otherwise it is $1 - L^{-(k+1)}$. Both are exactly the displayed expression. $\square$

Lemma 7.8 (The isolated-shift obstruction)**.**

`isolated_shift_not_tendsto_zero`

Let $k, Q > 0$, $A \in \mathbb{N}$, I a finite type, $r_i > 0$, and $c_i \in \mathbb{R}$. Suppose $i_0 \in I$, $c_{i_0} \neq 0$, and $r_i = r_{i_0}$ implies $i = i_0$. Then

$$\sum_{i \in I} c_i \rho_k(Qn + A + r_i) \not\longrightarrow 0.$$

Other shifts are allowed to coincide with one another.

Proof. Suppose this sum tends to zero. Choose L, v_0, v_1 by the separating residue lemma. For any v, the subsequence on the indices $Ln + v$ tends to zero, hence so does its Cesaro average. On the other hand, the progression-mean theorem with step QL and offset $Qv + A + r_i$, which is congruent to $A + r_i$ modulo Q, the prime-refinement factor, and finite weighted averaging give the limit

$$\sum_i c_i \mu_k(Q, A + r_i) \left(1 - L^{-(k+1)} + \mathbf{1}_{\{L \mid Qv+A+r_i\}} L^{-k}\right).$$

Hence this expression is zero for both v_0 and v_1. Subtract the two equalities. The first indicator misses every shift; the second selects exactly i_0. We obtain $c_{i_0} \mu_k(Q, A + r_{i_0}) L^{-k} = 0$. This is impossible: the first factor is nonzero, the mean is at least one (its $d = 1$ term is one and the others are nonnegative), and $L^{-k} > 0$. $\square$

8 Finite differences and the dilation grid

Define `diffWeight` by $w_o(a) = (-1)^{o-a}\binom{o}{a}$ for $0 \le a \le o$.

Lemma 8.1 (Vanishing polynomial moments)**.**

`diffWeight_moment`

If $\ell < o$ and $z, d \in \mathbb{R}$, then

$$\sum_{a=0}^{o} w_o(a)(z + da)^\ell = 0.$$

Proof. The sum is the order-o forward difference of x^ℓ, with step d, evaluated at z. Mathlib's iterated-forward-difference formula and its vanishing theorem for powers of degree below the order give zero. This also covers zero step and degree zero. $\square$

For a vertex $e \in \{0, \dots, o\}^n$, define `cubeWeight` by $W_o(e) = \prod_{a<n} w_o(e_a)$.

Lemma 8.2 (Cancellation in a coordinate absent from the argument)**.**

`cube_core`

Let $\ell < o$, $d, c : \{0, \dots, n-1\} \to \mathbb{R}$, $i < n$, $c_i = 0$, $g : \mathbb{R} \to \mathbb{R}$, and $p, q \in \mathbb{R}$. Then

$$\sum_{e \in \{0,\dots,o\}^n} W_o(e) \left(p + \sum_{a<n} d_a e_a\right)^\ell g\left(q + \sum_{a<n} c_a e_a\right) = 0.$$

Proof. Split each vertex into its coordinate e_i and the remaining coordinates. For fixed remaining coordinates the g factor and the other weight factors are independent of e_i, because $c_i = 0$. The sum over e_i is a fixed scalar times $\sum_{a=0}^{o} w_o(a)(z + d_i a)^\ell$, which vanishes by the moment lemma. Then sum over the remaining coordinates. Lean uses the coordinate-insertion equivalence directly; no induction over successive removed coordinates is needed. The dimension-zero case has no possible index i. $\square$

For fixed k, define the vertex type `GridVertex` and the constants `gridBound`, `gridSpacing`, and `gridBase` by

$$\mathcal{V}_k = \{0, \dots, k+1\}^k, \qquad b = k+2, \qquad F = b^k - 1, \qquad D = F!, \qquad B = 1 + kDF.$$

The definitions `gridIndex`, `gridWeightedIndex`, `gridMult`, `gridOffset`, and `gridShift` are

$$\begin{aligned} I_e &= \sum_{a<k} e_a b^a, & J_e &= \sum_{a<k} (a+1) e_a b^a, \\ p_e &= B + D I_e, & t_e &= D J_e, \\ s_{e,j} &= (j+1)p_e \dot{-} t_e. \end{aligned}$$

The zero vertex `gridZero` is $\mathbf{0} = (0, \dots, 0)$. The modulus `gridModulus` is $Q_k = \prod_{e \in \mathcal{V}_k} p_e^2$. These definitions also apply at $k = 0$, when the vertex type has one element.

Lemma 8.3 (The radix index bound)**.**

`gridIndex_le`

For every vertex, $I_e \le F$.

Proof. A length-k base-b digit string with digits below b represents an integer below b^k. In Lean this is the bound of Mathlib's finite radix-encoding equivalence. □

Lemma 8.4 (Uniqueness of radix encoding)**.**

`gridIndex_injective`

The map $e \mapsto I_e$ is injective.

Proof. Use uniqueness of the fixed-length base-b representation, formalized by the injectivity of the same radix-encoding equivalence. □

Lemma 8.5 (The zero index)**.**

`gridIndex_zero`

$I_{\mathbf{0}} = 0$.

Proof. Every summand is zero. □

Lemma 8.6 (The weighted index bound)**.**

`gridWeightedIndex_le`

For every vertex, $J_e \le kI_e$.

Proof. For each coordinate $a < k$, the factor $a+1$ is at most k. Multiply by $e_a b^a \ge 0$ and sum. □

Lemma 8.7 (Positive multipliers)**.**

`gridMult_pos`

$p_e > 0$ for every vertex and every k.

Proof. $p_e = B + DI_e$ and $B = 1 + kDF \ge 1$. □

Lemma 8.8 (Coprimality with the spacing)**.**

`gridMult_coprime_spacing`

$\gcd(p_e, D) = 1$.

Proof. $p_e = 1 + D(kF + I_e)$ is congruent to one modulo D. □

Lemma 8.9 (Coprimality for ordered indices)**.**

`multipliers_coprime_of_index_lt`

If $I_e < I_f$*, then* $\gcd(p_e, p_f) = 1$*.*

Proof. If a prime l divided both multipliers, it would divide $p_f - p_e = D(I_f - I_e)$. It is coprime to D, since it divides p_e; hence $l \mid I_f - I_e$. The latter integer is positive and at most F, so $l \le F$ and $l \mid F! = D$, a contradiction. No prime can divide both multipliers, so they are coprime. □

Lemma 8.10 (Pairwise coprime grid multipliers)**.**

`gridMult_pairwise_coprime`

Distinct vertices have coprime multipliers.

Proof. Their radix indices are distinct by injectivity. Order the two indices and apply the preceding lemma, using symmetry of coprimality if needed. □

Lemma 8.11 (The offsets are below the base)**.**

`gridOffset_lt_base`

$t_e < B$ *for every vertex.*

Proof. $t_e = DJ_e \le DkI_e \le DkF = B - 1$. □

Lemma 8.12 (All shifts are positive)**.**

`gridShift_pos`

$s_{e,j} > 0$ *for every vertex and every* j*.*

Proof. $t_e < B \le p_e \le (j+1)p_e$, so the natural difference is positive. □

Lemma 8.13 (Exact subtraction in the shifts)**.**

`gridShift_add_offset`

$s_{e,j} + t_e = (j+1)p_e$*.*

Proof. The preceding strict inequality guarantees that natural subtraction does not truncate. □

Lemma 8.14 (The unique distinguished shift)**.**

`gridShift_eq_succ_base_iff`

For $j \le k$*,*

$$s_{e,j} = (k+1)B \quad \iff \quad e = \mathbf{0} \quad \textit{and} \quad j = k.$$

Proof. If $j < k$, then $s_{e,j} \le k(B + DF) = kB + kDF = (k+1)B - 1$, so equality is impossible. Thus $j = k$. Now $s_{e,k} = (k+1)B + D((k+1)I_e - J_e)$, and $J_e \le kI_e$. Equality implies $DI_e = 0$; since $D > 0$, $I_e = 0$, and radix injectivity gives $e = \mathbf{0}$. Conversely, this vertex has $I_e = J_e = 0$, $p_e = B$, and $t_e = 0$, proving the equality at $j = k$. □

Lemma 8.15 (Positive CRT modulus)**.**

`gridModulus_pos`

$Q_k > 0$*.*

Proof. It is a finite product of the positive squares p_e^2. □

9 Cancellation and the final-order survivor

The integer weights `gridWeight` are $w_e = W_{k+1}(e) = \prod_{a<k}(-1)^{k+1-e_a}\binom{k+1}{e_a}$.

Lemma 9.1 (The real affine multiplier).

`gridMult_real_eq_cube`

Viewed as a real number, $p_e = B + \sum_{a<k} Db^a e_a$.

Proof. Expand $p_e = B + DI_e$ and distribute D through the sum. ☐

Lemma 9.2 (The real affine shift).

`gridShift_real_eq_cube`

For every j, viewed in $\mathbb{R}$,

$$s_{e,j} = (j+1)B + \sum_{a<k}(j-a)Db^a e_a.$$

Proof. Use the exact subtraction identity $s_{e,j} = (j+1)p_e - t_e$, substitute the sums for p_e and t_e, and combine the coefficients $(j+1)-(a+1) = j-a$. ☐

Lemma 9.3 (Cancellation on the actual integer grid).

`grid_core_cancel`

If $j < k$, $\ell \le k$, and $g : \mathbb{N} \to \mathbb{R}$, then

$$\sum_{e\in\mathcal{V}_k} w_e p_e^\ell g(s_{e,j}) = 0.$$

Proof. Use the two affine formulas in the cube-cancellation lemma with order $k+1$. Coordinate $a = j$ has coefficient zero in the shift, while the multiplier has degree $\ell < k+1$. To match the real-argument cube lemma exactly, extend g to $\mathbb{R}$ by $x \mapsto g(\max(\lfloor x \rfloor, 0))$. All actual shifts are natural numbers, so this extension agrees with g on every required argument. ☐

Define `gridCore`, `finiteMain`, and `gridCoeff` by

$$C_{k,N}(j,i) = \sum_{e\in\mathcal{V}_k} w_e p_e^{i+1}\frac{\sigma_k(N+s_{e,j})}{(N+s_{e,j})^{i+1}},$$

$$H_k(N) = \sum_{i=0}^{k}\sum_{j=0}^{k}\left\{ {i \atop j} \right\} C_{k,N}(j,i),$$

$$c_{e,j} = w_e p_e^{k+1}\left\{ {k \atop j} \right\}.$$

The definitions `survivingMain` and `survivor` are

$$S_k(N) = \sum_e\sum_{j=0}^{k} c_{e,j}\frac{\rho_k(N+s_{e,j})}{N+s_{e,j}}, \qquad V_k(N) = \sum_e\sum_{j=0}^{k} c_{e,j}\rho_k(N+s_{e,j}).$$

Lemma 9.4 (Only the top denominator order survives).

`finiteMain_eq_surviving`

For every k, N, $H_k(N) = S_k(N)$.

Proof. For an index $i < k$, if $j < k$, apply grid cancellation with $\ell = i+1 \le k$ and $g(s) = \sigma_k(N+s)/(N+s)^{i+1}$. The only other possible index is $j = k > i$, whose Stirling coefficient $\left\{ {i \atop j} \right\}$ is zero. Thus only $i = k$ remains. In that row use $\sigma_k(x)/x^{k+1} = \rho_k(x)/x$ and reorder the finite sums; the result is precisely $S_k(N)$. ☐

Lemma 9.5 (Regrouping the finite main term by vertices)**.**

`finiteMain_vertex`

For every k, N,

$$H_k(N) = \sum_e w_e \sum_{j=0}^{k} \sum_{i=0}^{k} \left\{ {i \atop j} \right\} p_e^{i+1} \frac{\sigma_k(N + s_{e,j})}{(N + s_{e,j})^{i+1}}.$$

Proof. Expand the core definition, distribute its coefficient, and commute the three finite sums. No convergence issue arises. □

Lemma 9.6 (The distinguished coefficient is nonzero)**.**

`gridCoeff_zero_ne_zero`

For every k, $c_{\mathbf{0},k} \neq 0$.

Proof. $\left\{ {k \atop k} \right\} = 1$, $p_{\mathbf{0}} = B > 0$, and each coordinate weight at zero is $(-1)^{k+1}$, hence nonzero. Their finite product, including the empty product at $k = 0$, is nonzero. □

Lemma 9.7 (The raw phase is sublinear)**.**

`tendsto_phase_div_nat`

For every k, $\rho_k(n)/n \longrightarrow 0$.

Proof. For $n > 0$, dividing the divisor bound by n^k gives $\rho_k(n) \leq 64\sqrt{n}$, so the ratio is between zero and $64\sqrt{n}/n$. The latter tends to zero by the square-root limit. □

Lemma 9.8 (Decay of each shifted main term)**.**

`grid_main_term_tendsto_zero`

For every k, e and every $h \in \mathbb{N}$,

$$c_{e,h} \frac{\rho_k(N + s_{e,h})}{N + s_{e,h}} \longrightarrow 0.$$

Proof. Apply the preceding limit along $N + s_{e,h} \to \infty$ and multiply by the fixed coefficient $c_{e,h}$. □

Lemma 9.9 (Decay of the complete surviving main term)**.**

`tendsto_survivingMain`

For every k, $S_k(N) \longrightarrow 0$.

Proof. Sum the finitely many preceding null sequences over e and $0 \leq j \leq k$. □

Lemma 9.10 (Rescaling to the raw survivor)**.**

`tendsto_survivor_rescaling`

For every k, $NS_k(N) - V_k(N) \longrightarrow 0$.

Proof. Every $s = s_{e,j}$ is positive, so $N + s \neq 0$. The corresponding difference is exactly

$$N\, c_{e,j} \frac{\rho_k(N + s)}{N + s} - c_{e,j}\rho_k(N + s) = -s\, c_{e,j} \frac{\rho_k(N + s)}{N + s}.$$

It tends to zero by the individual shifted-term limit. Sum these finitely many identities. □

10 CRT indices and the forced zero limit

Lemma 10.1 (A positive-step progression diverges)**.**

`tendsto_affine_atTop`

For $Q > 0$ and every A, $A + Qt \longrightarrow +\infty$.

Proof. $A + Qt \geq Qt$, and $Qt \to +\infty$. $\square$

The quotient index `tailIndex` is

$$n_e(N) = \left\lfloor \frac{N \dot{-} t_e}{p_e} \right\rfloor .$$

The property `GridCongruences`, denoted $\mathcal{C}_k(N)$, means that $N \equiv t_e \pmod{p_e^2}$ for every vertex e.

Lemma 10.2 (Exact common shifted arguments)**.**

`tailIndex_factorization`

If $N \geq t_e$ and $N \equiv t_e \pmod{p_e^2}$, then for every j,

$$p_e(n_e(N) + j + 1) = N + s_{e,j}.$$

Proof. The congruence implies $p_e \mid N - t_e$, so $p_e n_e(N) + t_e = N$. Add the exact shift identity $s_{e,j} + t_e = (j+1)p_e$ and rearrange. $\square$

Lemma 10.3 (Exact numerator and denominator rescaling)**.**

`tailIndex_term_rescale`

If $k > 0$, $j \leq k$, $N \geq t_e$, and $N \equiv t_e \pmod{p_e^2}$, then for every i,

$$\sigma_k(p_e) \frac{\sigma_k(n_e(N) + j + 1)}{(n_e(N) + j + 1)^{i+1}} = p_e^{i+1} \frac{\sigma_k(N + s_{e,j})}{(N + s_{e,j})^{i+1}}.$$

Proof. The squared congruence gives $p_e^2 \mid N \dot{-} t_e$, hence $p_e \mid n_e(N)$. Also $j + 1 \leq k + 1 \leq F$, since $b^k \geq b = k + 2$, so $j + 1 \mid F! = D$; as p_e is coprime to D, it is coprime to $j + 1$ and hence to $n_e(N) + j + 1$. This permits applying multiplicativity of σ_k to $p_e(n_e(N) + j + 1) = N + s_{e,j}$. Replace the denominator by the same product, distribute its power, and cancel $p_e^{i+1} \neq 0$. $\square$

Lemma 10.4 (Persistence along the CRT progression)**.**

`gridCongruences_add_modulus_mul`

If $\mathcal{C}_k(A)$, then $\mathcal{C}_k(A + Q_k t)$ for every $t \in \mathbb{N}$.

Proof. Each p_e^2 divides the product Q_k, so adding $Q_k t$ leaves every prescribed congruence unchanged. $\square$

The definitions `weightedTail`, `weightedMain`, and `weightedError` are respectively

$$W_T(N) = \sum_e w_e \sigma_k(p_e) T_k(n_e(N) + 1),$$
$$W_M(N) = \sum_e w_e \sigma_k(p_e) M_k(n_e(N)),$$
$$W_R(N) = \sum_e w_e \sigma_k(p_e) R_k(n_e(N)).$$

The dependence of these three expressions on k is suppressed.

Lemma 10.5 (The weighted exact decomposition)**.**

`weightedTail_split`

For every k, N, $W_T(N) = W_M(N) + W_R(N)$.

Proof. Apply $T_k(n+1) = M_k(n) + R_k(n)$ at each $n = n_e(N)$, multiply by $w_e\sigma_k(p_e)$, and sum. ☐

Lemma 10.6 (Divergence of every quotient index)**.**

`tendsto_tailIndex`

For every k, e, A and every $Q > 0$, $n_e(A + Qt) \longrightarrow +\infty$.

Proof. $A + Qt$ tends to infinity. Subtracting the fixed t_e, with eventual absence of truncation, still tends to infinity. Integer division by the positive constant p_e also tends to infinity. ☐

Lemma 10.7 (Eventual weighted integrality)**.**

`eventually_weightedTail_integral_prog`

If $\alpha_k \in \mathbb{Q}$, then for every A, $W_T(A + Q_k t) \in \mathbb{Z}$ eventually.

Proof. Every $n_e(A+Q_k t)+1$ tends to infinity, so the actual tail at that index is eventually integral. Intersect these finitely many eventual properties over the vertices. Each weight $w_e\sigma_k(p_e)$ is an integer, hence so is their weighted sum. No CRT assumption on A is needed for this lemma. ☐

Lemma 10.8 (Identifying the actual weighted main term)**.**

`weightedMain_eq_surviving`

If $k > 0$, $N \geq B$, and $\mathcal{C}_k(N)$, then $W_M(N) = S_k(N)$.

Proof. For every e, $t_e < B \leq N$. Expand $M_k(n_e(N))$ by its double Stirling sum and apply the exact term-rescaling lemma to each i, j. The regrouped expression is $H_k(N)$ by the vertex identity, and $H_k(N) = S_k(N)$ by cancellation. ☐

Lemma 10.9 (A rescaled vertex error)**.**

`tendsto_grid_vertex_error_mul`

If $\mathcal{C}_k(A)$, then for every e, along $N = A + Q_k t$,

$$NR_k(n_e(N)) \longrightarrow 0.$$

Proof. Eventually $N \geq t_e$, and the CRT factorization at $j = 0$ gives $N = p_e(n_e(N) + 1) - s_{e,0}$. Thus

$$NR_k(n_e(N)) = p_e(n_e(N) + 1)R_k(n_e(N)) - s_{e,0}R_k(n_e(N)).$$

Both terms on the right tend to zero: use the scaled and unscaled error limits at the diverging quotient index, with fixed coefficients p_e and $s_{e,0}$. This exact identity avoids a separate comparison bound. ☐

Lemma 10.10 (The rescaled weighted error)**.**

`tendsto_weightedError_mul`

If $\mathcal{C}_k(A)$, then along $N = A + Q_k t$, $NW_R(N) \longrightarrow 0$.

Proof. Distribute N through the finite sum for W_R. Each summand is the preceding null sequence multiplied by the fixed real coefficient $w_e\sigma_k(p_e)$. Signed coefficients cause no problem for a finite sum of convergent sequences. ☐

Lemma 10.11 (Rationality forces a zero survivor limit).

`tendsto_survivor_of_rational`

If $k > 0$, $\alpha_k \in \mathbb{Q}$, *and* $\mathcal{C}_k(A)$, *then* $V_k(A + Q_k t) \longrightarrow 0$.

Proof. Write $N = A + Q_k t$. First, $W_R(N) \to 0$ directly: each $R_k(n_e(N))$ tends to zero, and the weights form a fixed finite family. Also $W_M(N) = S_k(N)$ eventually, since eventually $N \geq B$ while all CRT congruences persist along the progression, and $S_k(N) \to 0$. The weighted decomposition therefore gives $W_T(N) \to 0$. It is eventually an integer by rationality, so it is eventually zero.

Consequently $S_k(N) = -W_R(N)$ eventually. Multiplying by N and using $NW_R(N) \to 0$ gives $NS_k(N) \to 0$. Finally $NS_k(N) - V_k(N) \to 0$ by the rescaling lemma. Subtracting this last null sequence yields $V_k(N) \to 0$. This is the combined limit argument in the current Lean proof; no separate unscaled weighted-error or weighted-tail limit lemmas remain. □

11 Irrationality in all degrees

Define `GridTerm` as $\mathcal{I}_k = \mathcal{V}_k \times \{0, \dots, k\}$. Its distinguished element is $(\mathbf{0}, k)$.

Lemma 11.1 (The survivor cannot tend to zero).

`survivor_not_tendsto_zero`

For $k > 0$, *every* $Q > 0$, *and every* $A \in \mathbb{N}$, $V_k(Qn + A) \not\longrightarrow 0$.

Proof. Apply the isolated-shift obstruction to the finite type $\mathcal{I}_k$, shifts $s_{e,j}$, and coefficients $c_{e,j}$. Every shift is positive. The distinguished shift is $(k+1)B$, occurs exactly once by the unique-shift lemma, and has nonzero coefficient by the distinguished-coefficient lemma. Regrouping the product-type sum is exactly the definition of V_k. □

Lemma 11.2 (Every positive degree is irrational).

`irrational_alpha_pos`

For $k > 0$, α_k *is irrational.*

Proof. Suppose it were rational. The squares p_e^2 are positive and pairwise coprime, so the finite Chinese remainder theorem gives A with $A \equiv t_e \pmod{p_e^2}$ for every vertex e. The forced-zero lemma would give $V_k(A + Q_k t) \to 0$. Since $Q_k > 0$, this contradicts the preceding obstruction, after commuting $A + Q_k t = Q_k t + A$. □

Lemma 11.3 (Positive actual tails).

`scaledTail_pos_of_pos`

For every k *and* $n > 0$, $T_k(n) > 0$.

Proof. The block expansion is summable and all its terms are nonnegative. Its first term is $B_k(n, 0) = \sigma_k(n)/n > 0$, since $n > 0$ and $\sigma_k(n) > 0$ (the divisor 1 already contributes one). The whole sum is at least this positive term. □

Lemma 11.4 (Decay of the degree-zero actual tail).

`tendsto_scaledTail_zero`

$T_0(n) \longrightarrow 0$.

Proof. At $k = 0$ the finite main term is $M_0(n) = \sigma_0(n+1)/(n+1) = \rho_0(n+1)/(n+1)$, which tends to zero by sublinearity of the phase. Also $R_0(n) \to 0$. Thus $T_0(n+1) = M_0(n) + R_0(n) \to 0$, and shifting the index back by one does not affect the limit. □

Lemma 11.5 (The divisor-count series is irrational)**.**

`irrational_alpha_zero`

α_0 *is irrational.*

Proof. If rational, $T_0(n)$ would be eventually integral. Its zero limit would make it eventually zero. This contradicts positivity for every $n > 0$. $\square$

Lemma 11.6 (The main theorem)**.**

`erdos_252`

For every $k \in \mathbb{N}$*,*

$$\sum_{n=0}^{\infty} \frac{\sum_{d \in \mathrm{divisors}(n)} d^k}{n!} \quad \textit{is irrational.}$$

Here $\mathrm{divisors}(0)$ *is empty; for* $n > 0$ *it is the set of positive divisors. The corresponding unchanged Lean statement is:*

```
theorem erdos_252 (k : ℕ) :
  Irrational (∑' n : ℕ,
    (ArithmeticFunction.sigma k n : ℝ) / (n.factorial : ℝ))
```

Proof. The displayed real number is the definition of α_k. For $k = 0$ use the degree-zero theorem; for $k > 0$ use the positive-degree theorem. $\square$

12 Statement audit and verification

This section uses the namespace `StatementAudit`. Its definition `publishedSum` is

$$\mathrm{publishedSum}(k) = \sum_{n \in \mathbb{N}}{}' \frac{\sigma_k(n)}{n!}.$$

The following six results are exactly those checked in `audit/Statement.lean`.

Lemma 12.1 (The positive-exponent published statement)**.**

`matches_published_statement`

For every $k \geq 1$*,* $\mathrm{publishedSum}(k)$ *is irrational.*

Proof. Apply `Erdos252.erdos_252`; its conclusion holds for every natural k, so in particular for $k \geq 1$. $\square$

Lemma 12.2 (The divisor-sum definition)**.**

`divisor_sum_definition`

For every k, n*,* $\mathrm{ArithmeticFunction.sigma}(k, n) = \sum_{d \in \mathrm{divisors}(n)} d^k$*.*

Proof. This is the defining equality of Mathlib's divisor-sum arithmetic function. $\square$

Lemma 12.3 (The zeroth summand)**.**

`zero_term`

For every k*,* $\sigma_k(0)/0! = 0$*.*

Proof. $\sigma_k(0) = 0$ and $0! = 1$. $\square$

Lemma 12.4 (Summability of the audited series).

`actual_series_summable`

For every k, $n \mapsto \sigma_k(n)/n!$ is summable.

Proof. Apply `Erdos252.summable_sigma_factorial`. ☐

Lemma 12.5 (The explicit divisor-set series).

`expanded_divisor_statement`

For every k, $\sum'_{n\in\mathbb{N}} (\sum_{d\in\mathrm{divisors}(n)} d^k)/n!$ is irrational.

Proof. Expand the divisor-sum definition in `Erdos252.erdos_252`. ☐

Lemma 12.6 (The equivalent positive-index series).

`positive_index_statement`

For every k, $\sum'_{n\in\mathbb{N}} \sigma_k(n+1)/(n+1)!$ is irrational.

Proof. Summability allows splitting the original series into its zeroth term and the shifted series. The zeroth term is zero, so the two series have the same sum. Apply the main theorem. ☐

Verification boundary and source

The unchanged main statement is proved in Lean 4.33.1 with mathlib revision

`0df444a360eaa60ab8c11dca51a86af692955474`

Its axiom dependencies are exactly `propext`, `Classical.choice`, and `Quot.sound`. The publication is checked by a full build, a fresh trust-zero build, the statement and axiom audits, and a fresh replay with Lean's kernel. The replay uses Lean's own kernel, not a separate kernel implementation. The prose proofs in this PDF have been checked against the named Lean results, but are not themselves kernel input.

The complete Lean source is the single module `Erdos252/Solution.lean`. The repository README contains reproduction commands, and the statement audit records the six checks above. See also the original problem statement.

This edition covers all 94 named results and 40 definitions or abbreviations in the proof module, plus the six audit results and their one definition. The proof-module SHA-256 (UTF-8, LF line endings) is

`601c377d7a063ef7b5360ef7caca7216ae982406d194571be463dd589b8b53c5`